



1. Description

1.1. Project

Project Name	equipo_medicion_ensayo_celdas
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	08/16/2026

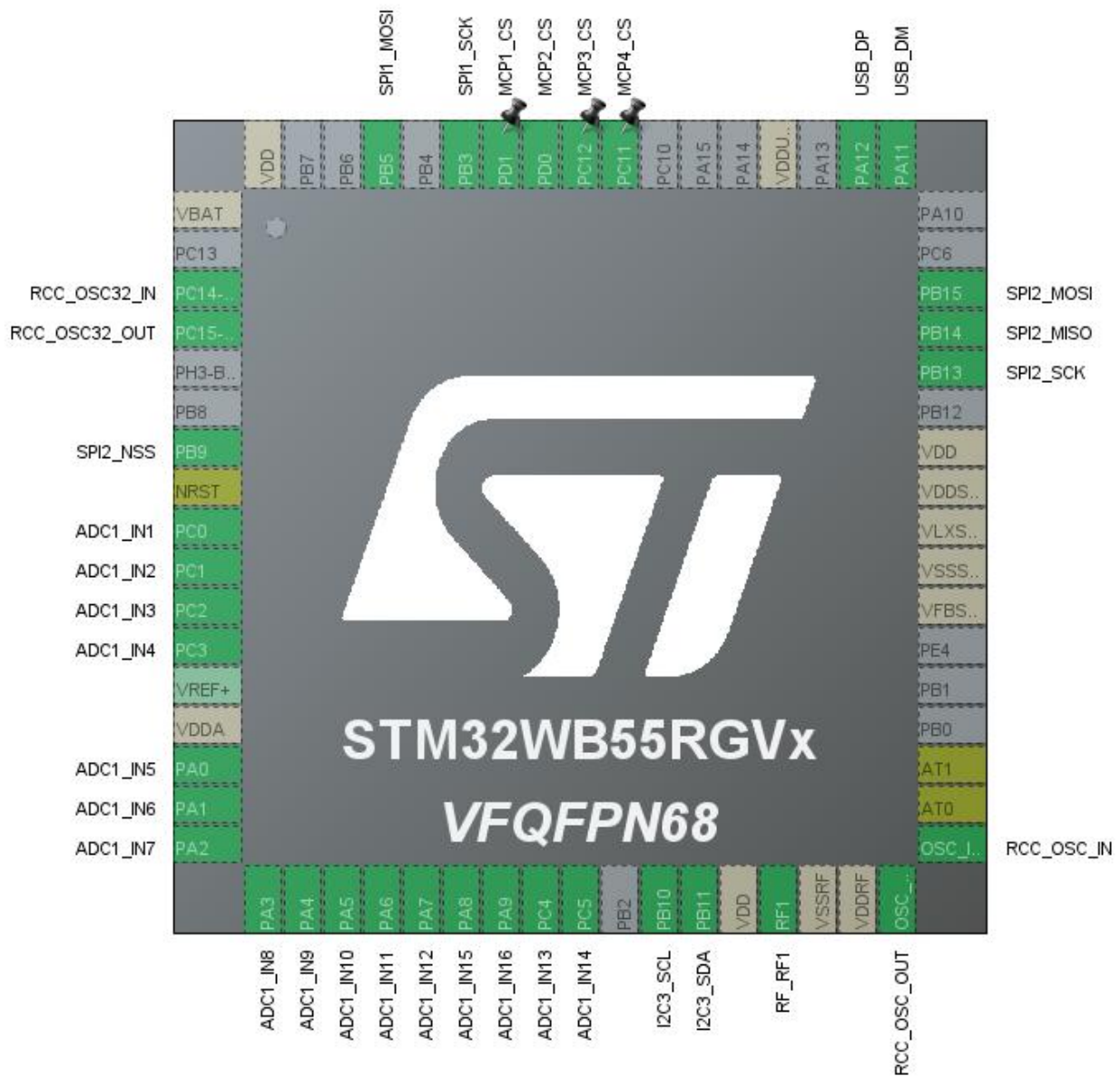
1.2. MCU

MCU Series	STM32WB
MCU Line	STM32WBx5
MCU name	STM32WB55RGVx
MCU Package	VFQFPN68
MCU Pin number	68

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

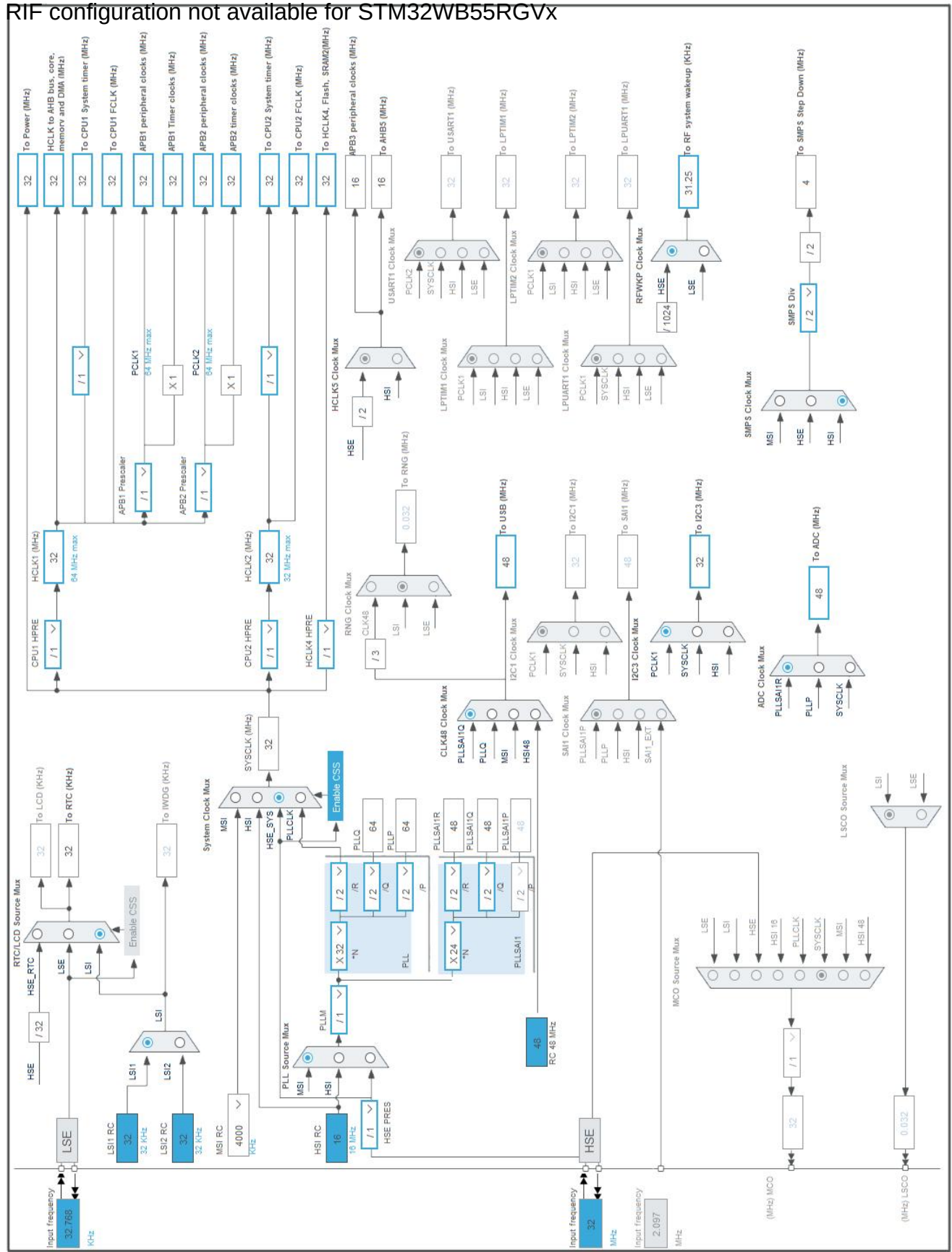
Pin Number VFQFPN68	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
3	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
4	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
7	PB9	I/O	SPI2_NSS	
8	NRST	Reset		
9	PC0	I/O	ADC1_IN1	
10	PC1	I/O	ADC1_IN2	
11	PC2	I/O	ADC1_IN3	
12	PC3	I/O	ADC1_IN4	
14	VDDA	Power		
15	PA0	I/O	ADC1_IN5	
16	PA1	I/O	ADC1_IN6	
17	PA2	I/O	ADC1_IN7	
18	PA3	I/O	ADC1_IN8	
19	PA4	I/O	ADC1_IN9	
20	PA5	I/O	ADC1_IN10	
21	PA6	I/O	ADC1_IN11	
22	PA7	I/O	ADC1_IN12	
23	PA8	I/O	ADC1_IN15	
24	PA9	I/O	ADC1_IN16	
25	PC4	I/O	ADC1_IN13	
26	PC5	I/O	ADC1_IN14	
28	PB10	I/O	I2C3_SCL	
29	PB11	I/O	I2C3_SDA	
30	VDD	Power		
31	RF1	MonoIO	RF_RF1	
32	VSSRF	Power		
33	VDDRF	Power		
34	OSC_OUT	MonoIO	RCC_OSC_OUT	
35	OSC_IN	MonoIO	RCC_OSC_IN	
36	AT0	NC		
37	AT1	NC		
41	VFBSMPS	Power		
42	VSSSMPS	Power		
43	VLXSMPS	Power		
44	VDDSMPS	Power		

Pin Number VFQFPN68	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
45	VDD	Power		
47	PB13	I/O	SPI2_SCK	
48	PB14	I/O	SPI2_MISO	
49	PB15	I/O	SPI2_MOSI	
52	PA11	I/O	USB_DM	
53	PA12	I/O	USB_DP	
55	VDDUSB	Power		
59	PC11 *	I/O	GPIO_Output	MCP4_CS
60	PC12 *	I/O	GPIO_Output	MCP3_CS
61	PD0 *	I/O	GPIO_Output	MCP2_CS
62	PD1 *	I/O	GPIO_Output	MCP1_CS
63	PB3	I/O	SPI1_SCK	
65	PB5	I/O	SPI1_MOSI	
68	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32WB55RGVx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32WB
Line	STM32WBx5
MCU	STM32WB55RGVx
Datasheet	DS11929 Rev3

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

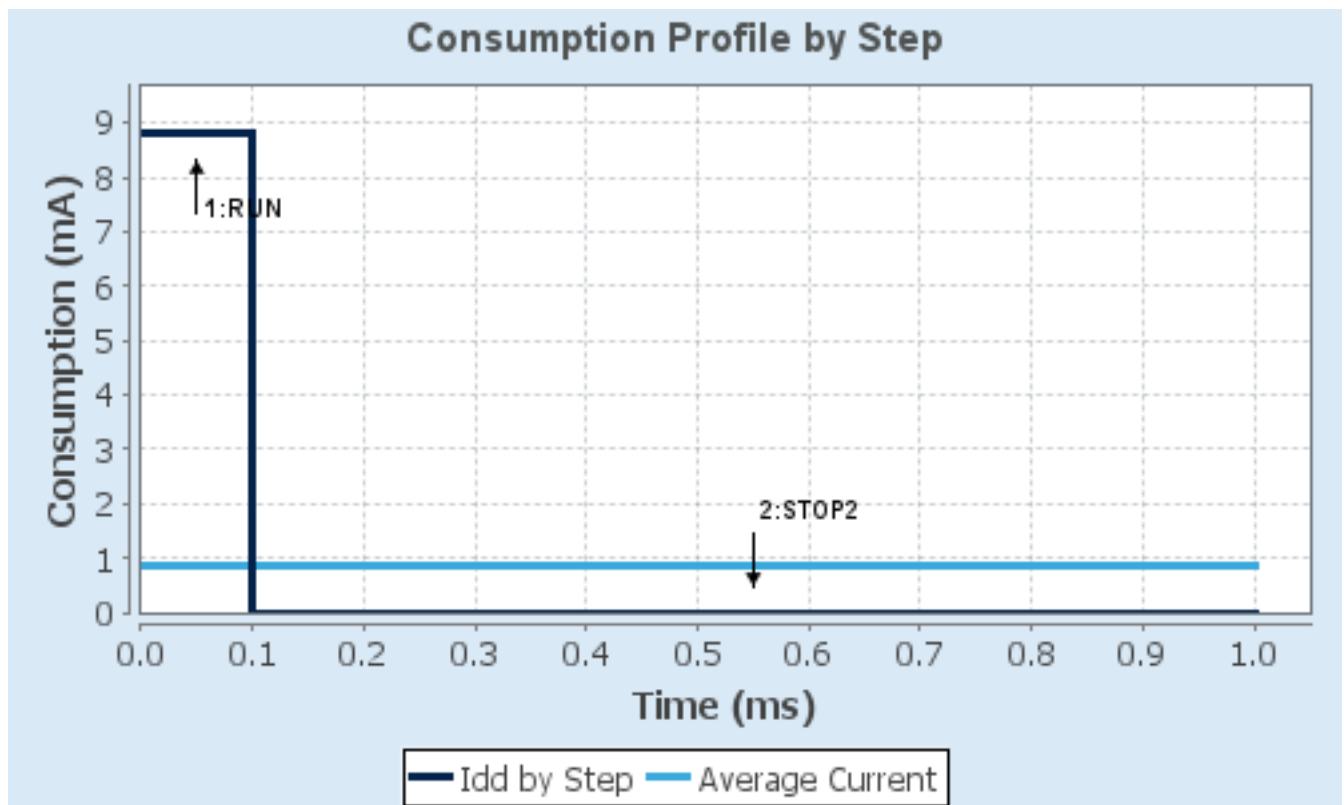
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP2
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-High	NoRange
Fetch Type	SRAM1/Flash-PowerDown	FLASH/ART/CACHE
CPU Frequency	64 MHz	0 Hz
Clock Configuration	HSI PLL Regulator_ON	ALL CLOCKS OFF Regulator_ON
Clock Source Frequency	16 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	8.8 mA	1.85 μ A
Duration	0.1 ms	0.9 ms
DMIPS	80.0	0.0
Ta Max	103.76	105
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	881.66 μ A
Battery Life	5 months, 7 days, 21 hours	Average DMIPS	8.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	equipo_medicion_ensayo_celdas
Project Folder	C:\Users\German\Documents\equipo_medicion_ensayo_celdas\firmware
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_WB V1.24.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ADC1_Init	ADC1
4	MX_I2C3_Init	I2C3
5	MX_IPCC_Init	IPCC
6	MX_RTC_Init	RTC
7	MX_SPI1_Init	SPI1
8	MX_SPI2_Init	SPI2
9	MX_USB_Device_Init	USB_DEVICE
10	APPE_Init	STM32_WPAN

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN2: IN2 Single-ended

IN3: IN3 Single-ended

IN4: IN4 Single-ended

IN5: IN5 Single-ended

IN6: IN6 Single-ended

IN7: IN7 Single-ended

IN8: IN8 Single-ended

IN9: IN9 Single-ended

IN10: IN10 Single-ended

IN11: IN11 Single-ended

IN12: IN12 Single-ended

IN13: IN13 Single-ended

IN14: IN14 Single-ended

IN15: IN15 Single-ended

mode: IN16 Single-ended

3.1.1. Parameter Settings:

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1

Channel	Channel 1
Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.2. HSEM

mode: Activated

3.3. I2C3

I2C: I2C

3.3.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00B07CB4 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.4. IPCC

mode: Activated

3.5. MEMORYMAP

mode: Activated

3.6. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.6.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	1 WS (2 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
MSI Calibration Value	0
MSI Auto Calibration	Enabled
MSI State	Enabled
HSI State	Enabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
LSE Drive Capability	LSE oscillator medium high drive capability

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE
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3.7. RF

mode: Activate RF1

3.8. RTC

mode: Activate Clock Source

WakeUp: Internal WakeUp

3.8.1. Parameter Settings:

General:

Hour Format	Hourformat 24
Asynchronous Predivider value	CFG_RTC_ASYNC_PRESCALER
Synchronous Predivider value	CFG_RTC_SYNC_PRESCALER

Wake UP:

Wake Up Clock	1 Hz *
Wake Up Counter	0

3.9. SPI1

Mode: Transmit Only Master

3.9.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	16.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

3.10. SPI2

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.10.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits

First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	16.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Output Hardware

3.11. SYS

Timebase Source: TIM17

3.12. TINY_LPM

mode: Enabled

3.13. USB

mode: Device (FS)

3.13.1. Parameter Settings:

Basic Parameters:	
Speed	Full Speed 12MBit/s
Physical interface	Internal Phy
Sof Enable	Disabled
Power Parameters:	
Low Power	Disabled
Link Power Management	Disabled
Battery Charging	Disabled

3.14. FREERTOS

Interface: CMSIS_V2

3.14.1. Config parameters:

API:	
FreeRTOS API	CMSIS v2

Versions:

FreeRTOS version	10.6.2
CMSIS-RTOS version	2.1.3

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
USE_SB_COMPLETED_CALLBACK	0
USE_MINI_LIST_ITEM	1
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled
OVERRIDE_DEFAULT_TICK_CONFIGURATION	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	20480 *
HEAP_CLEAR_MEMORY_ON_FREE	0
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled

USE_STATS_FORMATTING_FUNCTIONS Disabled

Co-routine related definitions:

USE_CO_ROUTINES Disabled

MAX_CO_ROUTINE_PRIORITIES 2

Software timer definitions:

USE_TIMERS Enabled

TIMER_TASK_PRIORITY 2

TIMER_QUEUE_LENGTH 10

TIMER_TASK_STACK_DEPTH 256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY 15

LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY 5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE size_t

USE_POSIX_ERRNO Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME Enabled

USE_OS2_THREAD_ENUMERATE Enabled

USE_OS2_EVENTFLAGS_FROM_ISR Enabled

USE_OS2_THREAD_FLAGS Enabled

USE_OS2_TIMER Enabled

USE_OS2_MUTEX Enabled

3.14.2. Include parameters:

Include definitions:

vTaskPrioritySet Enabled

uxTaskPriorityGet Enabled

vTaskDelete Enabled

vTaskCleanUpResources Disabled

vTaskSuspend Enabled

vTaskDelayUntil Enabled

vTaskDelay Enabled

xTaskGetSchedulerState Enabled

xTaskResumeFromISR Enabled

xQueueGetMutexHolder Enabled

xSemaphoreGetMutexHolder Disabled

pcTaskGetTaskName Disabled

uxTaskGetStackHighWaterMark Enabled

xTaskGetCurrentTaskHandle Enabled

eTaskGetState Enabled

xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.14.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
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Project settings (see parameter description first):

Use FW pack heap file	Enabled
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3.15. STM32_WPAN

mode: BLE

3.15.1. BLE Applications and Services:

BLE Wireless Stack:

BLE Wireless Stack	Full
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BLE Application Type:

BLE Application Type	Server profile
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Server Mode:

BT SIG Heart Rate Sensor	Disabled *
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BLE Services Configuration:

The device needs to support the Peripheral Role	1
The device needs to support the Central Role	0
BLE_CFG_SVC_MAX_NBR_CB	7
BLE_CFG_CLT_MAX_NBR_CB	0

3.15.2. Configuration:

HW Timer Server:

CFG_HW_TS_MAX_NBR_CONCURRENT_TIMER	6
CFG_HW_TS_NVIC_RTC_WAKEUP_IT_PREEMPTPRIO	3
CFG_HW_TS_NVIC_RTC_WAKEUP_IT_SUBPRIO	0
CFG_HW_TS_USE_PRIMASK_AS_CRITICAL_SECTION	1
CFG_HW_TS_RTC_HANDLER_MAX_DELAY	(10 * (LSI_VALUE/1000))
CFG_HW_TS_RTC_WAKEUP_HANDLER_ID	RTC_WKUP_IRQn

HW UART:

CFG_HW_LPUART1_ENABLED	Disabled
CFG_HW_LPUART1_DMA_TX_SUPPORTED	Disabled
CFG_HW_USART1_ENABLED	Disabled
CFG_HW_USART1_DMA_TX_SUPPORTED	Disabled

Generic parameters:

CFG_HW_RESET_BY_FW	Disabled
CFG_USE_SMPS	Disabled
CFG_LPM_SUPPORTED	Disabled
CFG_DEBUGGER_SUPPORTED	Enabled
CFG_DEBUG_BLE_TRACE	Disabled
CFG_DEBUG_APP_TRACE	Disabled
CFG_DEBUG_TRACE_LIGHT	Disabled
CFG_DEBUG_TRACE_FULL	Disabled
DBG_TRACE_USE_CIRCULAR_QUEUE	Enabled
DBG_TRACE_MSG_QUEUE_SIZE	4096
MAX_DBG_TRACE_MSG_SIZE	1024

Application parameters:

CFG_TX_POWER	-0.15dBm (0x18)
CFG_DEBUG_TRACE_UART	You need to activate either CFG_HW_UART1 or CFG_HW_LPUART1(when available)
CFG_CONSOLE_MENU	You need to activate either CFG_HW_UART1 or CFG_HW_LPUART1(when available)
CFG_ADV_BD_ADDRESS	0x11aabbccdde *
CFG_FAST_CONN_ADV_INTERVAL_MIN	80
CFG_FAST_CONN_ADV_INTERVAL_MAX	100
CFG_LP_CONN_ADV_INTERVAL_MIN	1000
CFG_LP_CONN_ADV_INTERVAL_MAX	2500
CFG_IO_CAPABILITY	Display Yes No (0x01)
CFG_MITM_PROTECTION	MITM protection required (0x01)
CFG_RTCCLK_DIVIDER_CONF	0
CFG_RTCCLK_DIV	16
CFG_RTC_WUCKSEL_DIVIDER	0
CFG_RTC_ASYNC_PRESCALER	0x0F *
CFG_RTC_SYNC_PRESCALER	0x7FFF *
CFG_BLE_NUM_LINK	2
CFG_BLE_NUM_GATT_SERVICES	8
CFG_BLE_NUM_GATT_ATTRIBUTES	68
CFG_BLE_MAX_ATT_MTU	156
CFG_BLE_ATT_VALUE_ARRAY_SIZE	1344
CFG_BLE_DATA_LENGTH_EXTENSION	Enabled

CFG_BLE_PERIPHERAL_SCA	500
CFG_BLE_CENTRAL_SCA	0
CFG_BLE_HSE_STARTUP_TIME	0x148 *
CFG_BLE_MAX_CONN_EVENT_LENGTH	0xFFFFFFFF *
CFG_BLE_VITERBI_MODE	Enabled
CFG_BLE_OPTIONS	BLE stack Options flags:
- CFG_BLE_OPTIONS_LL	SHCI_C2_BLE_INIT_OPTIONS_LL_HOST
- CFG_BLE_OPTIONS_SVC	SHCI_C2_BLE_INIT_OPTIONS_WITH_SVC_CHANGE_DESC
- CFG_BLE_OPTIONS_DEVICE_NAME	SHCI_C2_BLE_INIT_OPTIONS_DEVICE_NAME_RW
- CFG_BLE_OPTIONS_EXT_ADV	SHCI_C2_BLE_INIT_OPTIONS_NO_EXT_ADV
- CFG_BLE_OPTIONS_CS_ALGO	SHCI_C2_BLE_INIT_OPTIONS_NO_CS_ALGO2
- CFG_BLE_OPTIONS_GATTDDB_NVM	SHCI_C2_BLE_INIT_OPTIONS_FULL_GATTDDB_NVM
- CFG_BLE_OPTIONS_GATT_CACHING	SHCI_C2_BLE_INIT_OPTIONS_GATT_CACHING_NOTUSED
- CFG_BLE_OPTIONS_POWER_CLASS	SHCI_C2_BLE_INIT_OPTIONS_POWER_CLASS_2_3
- CFG_BLE_OPTIONS_APPEARANCE	SHCI_C2_BLE_INIT_OPTIONS_APPEARANCE_READONLY
- CFG_BLE_OPTIONS_ENHANCED_ATT	SHCI_C2_BLE_INIT_OPTIONS_ENHANCED_ATT_NOTSUPPORTED
CFG_BLE_MAX_COC_INITIATOR_NBR	32
CFG_BLE_MIN_TX_POWER	0
CFG_BLE_MAX_TX_POWER	0
CFG_BLE_MAX_ADD_EATT_BEARERS	4
CFG_BLE_RX_MODEL_CONFIG	SHCI_C2_BLE_INIT_RX_MODEL_AGC_RSSI_LEGACY
CFG_BLE_MAX_ADV_SET_NBR	3
CFG_BLE_MAX_ADV_DATA_LEN	1650
CFG_BLE_TX_PATH_COMPENS	0
CFG_BLE_RX_PATH_COMPENS	0
CFG_BLE_CORE_VERSION	SHCI_C2_BLE_INIT_BLE_CORE_5_4
CFG_TLBLE_EVT_QUEUE_LENGTH	5
CFG_TLBLE_MOST_EVENT_PAYLOAD_SIZE	255
Pairing parameters:	
CFG_BONDING_MODE	No-bonding mode(0x00)
CFG_ENCRYPTION_KEY_SIZE_MAX	16
CFG_ENCRYPTION_KEY_SIZE_MIN	8
CFG_SC_SUPPORT	Secure Connections Pairing supported but optional (0x01)

CFG_BLE_IR	12, 34, 56, 78, 9A, BC, DE, F0, 12, 34, 56, 78, 9A, BC, DE, F0
CFG_BLE_ER	FE, DC, BA, 09, 87, 65, 43, 21, FE, DC, BA, 09, 87, 65, 43, 21
CFG_KEYPRESS_NOTIFICATION_SUPPORT	Keypress notification not supported (0x00)
Debug options:	
BLE_DBG_APP_EN	Disabled

3.16. USB_DEVICE

Class For FS IP: Communication Device Class (Virtual Port Com)

3.16.1. Parameter Settings:

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message
USBD_LPM_ENABLED (Link Power Management)	1: Link Power Management supported

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

3.16.2. Device Descriptor:

Device Descriptor:

VID (Vendor Identifier)	1155
LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	STMicroelectronics

Device Descriptor FS:

PID (Product Identifier)	22336
PRODUCT_STRING (Product Identifier)	STM32 Virtual ComPort
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface

* User modified value

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	
	PC1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	
	PC2	ADC1_IN3	Analog mode	No pull-up and no pull-down	n/a	
	PC3	ADC1_IN4	Analog mode	No pull-up and no pull-down	n/a	
	PA0	ADC1_IN5	Analog mode	No pull-up and no pull-down	n/a	
	PA1	ADC1_IN6	Analog mode	No pull-up and no pull-down	n/a	
	PA2	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	
	PA3	ADC1_IN8	Analog mode	No pull-up and no pull-down	n/a	
	PA4	ADC1_IN9	Analog mode	No pull-up and no pull-down	n/a	
	PA5	ADC1_IN10	Analog mode	No pull-up and no pull-down	n/a	
	PA6	ADC1_IN11	Analog mode	No pull-up and no pull-down	n/a	
	PA7	ADC1_IN12	Analog mode	No pull-up and no pull-down	n/a	
	PA8	ADC1_IN15	Analog mode	No pull-up and no pull-down	n/a	
	PA9	ADC1_IN16	Analog mode	No pull-up and no pull-down	n/a	
	PC4	ADC1_IN13	Analog mode	No pull-up and no pull-down	n/a	
	PC5	ADC1_IN14	Analog mode	No pull-up and no pull-down	n/a	
I2C3	PB10	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB11	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
	OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
RF	RF1	RF_RF1	n/a	n/a	n/a	
SPI1	PB3	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB5	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SPI2	PB9	SPI2_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB13	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USB	PA11	USB_DM	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA12	USB_DP	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PC11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	MCP4_CS

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	MCP3_CS
	PD0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	MCP2_CS
	PD1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	MCP1_CS

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
USB low priority interrupt, USB wake-up interrupt through EXTI line 28	true	5	0
TIM1 trigger and commutation interrupts and TIM17 global interrupt	true	15	0
IPCC RX occupied interrupt	true	5	0
IPCC TX free interrupt	true	5	0
HSEM global interrupt	true	5	0
PVD/PVM0/PVM2 interrupts through EXTI lines 16/31/33	unused		
RTC wake-up interrupt through EXTI line 19	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 global interrupt	unused		
USB high priority interrupt	unused		
CPU2 SEV interrupt through EXTI line 40 and PWR CPU2 HOLD wake-up interrupt	unused		
I2C3 event interrupt	unused		
I2C3 error interrupt	unused		
SPI1 global interrupt	unused		
SPI2 global interrupt	unused		
PWR switching on the fly, end of BLE activity, end of 802.15.4 activity, end of critical radio phase interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
USB low priority interrupt, USB wake-up interrupt through EXTI line 28	false	true	true
TIM1 trigger and commutation interrupts and TIM17 global interrupt	false	true	true
IPCC RX occupied interrupt	false	true	true
IPCC TX free interrupt	false	true	true
HSEM global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware								
FREERTOS STM32_WPAH USB_DEVICE								
System Core	Analog	Timers	Connectivity	Multimediac	Security	Computing	Utilities	Other
DMA	ADC1	RTC	I2C3				TINY_LPM	
GPIO			RF					
HSEM			SPH1					
IPCC			SPI2					
IVIC			USB					
RCC								
SYS								

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32wb_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32wb_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32wb_svd.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/stm32wb55rg_mb1479_ref_board.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32wbxm_wireless-modules_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-graphics-solution-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-graphics-solutions-detailed.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wb.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wbvl.pdf

Flyers	https://www.st.com/resource/en/flyer/flstm32matter.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wbxm.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32zigbee.pdf
White Papers	https://www.st.com/resource/en/white_paper/seamless-smart-home-connectivity-with-matter-whitepaper.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/stm32wb-rf-certificates.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/ble-thread-ftd-dynamic-thread-device-interoperability-certificate.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/full-thread-device-interoperability-certification.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/minimal-thread-device-interoperability-certification.pdf
Security Advisory	https://www.st.com/resource/en/security_advisory/sa0024-potential-isolation-issue-between-cpu1-and-cpu2-on-stm32wb5x-stm32wb3x-stm32wb1x-and-stm32wl5x-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5071-stm32wb-series-microcontrollers-ultralowpower-features-overview-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5492-persistent-data-management-zigbee-and-nonvolatile-memory-in-stm32wb-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5498-how-to-use-zigbee-clusters-templates-on-stm32wb-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5500-zsdk-api-implementation-for-zigbee-on-stm32wb-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5506-getting-started-with-zigbee-on-stm32wb-series-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5627-stm32wb-series-zigbee-commissioning-guide-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5707-st-bluetooth-mesh-sensor-model-stmicroelectronics.pdf

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- Application Notes https://www.st.com/resource/en/application_note/an5105-getting-started-with-touch-sensing-control-on-stm32-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5395-how-to-drive-an-external-power-amplifier-with-stm32wb-series-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5155-stm32cube-mcu-package-examples-for-the-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-how-to-use-uart-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-how-to-use-usb-dfu-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-how-to-use-spi-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an4776-how-to-use-generalpurpose-timer-peripheral-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5247-guidelines-for-overtheair-application-and-wireless-firmware-update-for-stm32wb-series-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5434-guidelines-for-onboard-antenna-reference-designs-for-stm32wbwba-mcus-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0394-stm32wb55xxstm32wb35cx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00344191.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf

Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0223-stm32-cortexm0-mcus-programming-manual-stmicroelectronics.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0271-guidelines-for-bluetooth-le-stack-programming-on-stm32wbstm32wba-mcus-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0434-multiprotocol-wireless-32bit-mcu-armbased-cortexm4-with-fpu-bluetooth-lowenergy-and-802154-radio-solution-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-devices-in-bga-packages-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2804-stm32wb-series-ble-low-level-driver-ll-d-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2977-stm32wb-series-zigbee-cluster-library-api-stmicroelectronics.pdf